

Recitation 9 Spring 2023

Hashing

1. Answer the following and explain

- I. What is the time complexity of insertion in a hash table?
- II. What is the time complexity of searching in a hash table?
- III. What is the time complexity of deletion in a hash table?
- IV. Is chained hashing best used when there is a low load factor or high load factor?
- V. Is linear probing best used when there is a low load factor or high load factor?

2. Insert the following numbers into this hash table below, using the specified probing method and the given hash function.

$$h(k) = (2k) \bmod 11$$

Values = {19, 65, 09, 17, 01, 99, 12, 4, 6, 23, 22}

Linear Probing:

[illegible]

Quadratic Probing:

[illegible]

3. Given a hash table with size 48 with 12 elements in it, what is the load factor? Is it possible to have more than 48 elements in this hash table?

4. Consider the following hash function:

$$h(k) = k \bmod 4$$

Values = {33, 17, 9, 40, 44, 59, 12, 19, 38}

Which values will have $h = 0$? $h = 1$? $h = 2$? $h = 3$?

Is this an efficient hash function? Explain why or why not.

5. Consider a hash table of size 11, done via double hashing with the following hash functions:

$$h_1(k) = k \bmod 11$$

$$h_2(k) = 5 - (k \bmod 5)$$

Values = {32, 43, 26, 38, 17, 3, 80, 55}

Draw the table that results after inserting, in the given order, the values. **(10 minutes)**

Hash Table Index	0	1	2	3	4	5	6	7	8	9	10
Hash Table Entries											

6. A hash table is defined using an array of 50 IntNode references such that the collisions can be handled by using chaining, as follows:

```
public class Table {
    private int manyItems;
    private IntNode[] keys;

    public Table() {
        keys = new IntNode[50];
        manyItems = 0;
    }
    private int hash(int key) {
        return key % 100; } }

```

Write Java code for the following hash Table methods below. You may assume that IntNode class has all of its standard methods.

a) public void put (int key) //Inserts key into the appropriate “chain” of the hash table
//You may assume that the key isn’t a duplicate

b) public boolean containsKey(int key)
//Returns true if key is in the hash table, otherwise it returns false.

7. Given an array of integers, find the mode of the array using a hash map ()
public int mode(int[] arr) {

